CHRIS SKINNER'S COURSE FALL 2020

GYUJIN OH

1. Lecture 01, 8/31

Our focus is on special cycles and classes in cohomology of Galois representations of automorphic forms and their applications to bounding Selmer groups, and Iwasawa theory.

A motivation comes from Birch–Swinnerton-Dyer conjecture. Let $E/\mathbb{Q}$ be an elliptic curve, $E : y^2 = x^3 + Ax + B$ of nonzero discriminant, with $A, B \in \mathbb{Q}$. We are interested about rational points of E (over $\mathbb{F}_p$, $\mathbb{Q}_p$, $\mathbb{Q}$).

- Over $\mathbb{F}_p$: if $p \nmid 2\Delta$ then reduction mod p is elliptic curve over $\mathbb{F}_p$. The Riemann hypothesis says that $\#E(\mathbb{F}_p) = 1+p-a_p$ where $|a_p| \leq 2\sqrt{p}$. Even better, the roots of $x^2-a_px+p = 0$ has absolute values $p^{1/2}$.
- Over $\mathbb{Q}$: by Mordell, $E(\mathbb{Q})$ is finitely generated abelian group. Even the list of possible torsion parts is completely understood by Mazur. In general, torsion points are very rare (0% of elliptic curves have a torsion point).

  On the other hand, the rank of an elliptic curve is mysterious.
  - How large the rank can be? We don't really know. The current trend is that it's probably bounded. There was a time where people thought it's not.
  - Goldfeld/Katz–Sarnak conjecture expects that half of the time rank is zero and half of the time rank is one.

  Consider the Hasse-Weil L-function $L(s, E) = \prod_p L_p(p^{-s})^{-1}$ where $L_p(x) \in \mathbb{Q}[x]$ is a polynomial of degree ≤ 2 and if E has good reduction at p then $L_p(x) = 1 - a_px + px^2$. The Riemann hypothesis we alluded above implies that $L(s, E)$ converges absolutely for $\mathrm{Re}(s) > \frac{3}{2}$.

  Conjecture 1.1 (Hasse–Weil). *$L(s, E)$ has an analytic continuation to $s \in \mathbb{C}$ and there is a functional equation $s \leftrightarrow 2 - s$.*

  A CM elliptic curve is an elliptic curve E where, if $E(\mathbb{C}) \cong \mathbb{C}/\Lambda$, then $\mathrm{End}(E) = \{\alpha \in \mathbb{C} \mid \alpha\Lambda \subset \Lambda\}$ is an order in an imaginary quadratic field. Theorem of CM (Deuring) says that $L(s, E) = L(s, \psi)$ for a Hecke character ψ of Hecke character of CM field. Thus this provides analytic continuation and functional equation.

  In general, for all curves, it needs modularity of elliptic curves.

  These are related to concrete arithmetic problems.

  (1) Congruent number problem: a positive number n is **congruent** if it is an area of a right triangle with side lengths in $\mathbb{Q}$. We know that n is congruent if and only if the rank of $E_n : y^2 = x^2 - n^2x$ has positive rank. E_n has CM by $\mathbb{Q}(\sqrt{-1})$.

  (2) The problem of whether n is a sum of two cubes is related to $E^{(n)} : y^2 = x^3 - 432n^2$. Namely it is equivalent to $E^{(n)}$ having positive rank. Note that $E^{(n)}$ has CM by $\mathbb{Q}(\sqrt{-3})$.

The BSD conjecture attempts to answer when $r_E > 0$. Let $N_p = \#E(\mathbb{F}_p) = 1 + p - a_p$. It is $O(p)$ so let's consider $\frac{N_p}{p}$. If there are many rational points then there might be more N_p than you expect. Birch–Swinnerton-Dyer computed and discovered that

$$\prod_{p \le X} \frac{N_p}{p} \sim c_E (\log X)^{r_E}$$

(they did a calculation for CM elliptic curves actually).

Note that $\frac{N_p}{p} = L_p(p^{-1})$. So $L(1, E)$ "looks like" $\left(\prod_p \frac{N_p}{p}\right)^{-1}$, which converts into the BSD conjecture:

Conjecture 1.2 (BSD)**.** $r_E = \operatorname{ord}_{s=1} L(s, E)$.

Note that it requires that $L(s, E)$ can be first continued to region around $s = 1$. They went further:

Conjecture 1.3 (Refined BSD)**.**

$$\frac{L^{(r_E)}(1, E)}{r_E! \Omega_E \operatorname{reg} E} = \frac{\#Ш(E) \prod_p c_p(E)}{(\#E(\mathbb{Q})_{\text{tors}})^2}$$

We will explain what these notations are.

Remark 1.1. Goldfeld showed that the asymptotic

$$\prod \frac{N_p}{p} \sim c_E (\log X)^{r_E}$$

implies GRH for $L(E, s)$, which implies BSD.

We now explain the parts of refined BSD conjecture.

- $\Omega_E = \int_{E(\mathbb{R})} \omega_E$ (**(real) period**), where ω_E is the Neron differential of E, which is a normalized basis of the (1-dimensional) space of holomorphic 1-forms coming from the Neron model of E.
- $\operatorname{reg} E$ (**regulator**) is the determinant of the Neron–Tate height pairing.
- $Ш(E)$ (**Tate–Shafarevich group**) has many descriptions.
  - It is the group of isomorphism classes of $\mathbb{Q}$-forms of E that are trivial over all local fields $\mathbb{Q}_p$ for $p \le \infty$.
  - $$Ш(E) = \ker\{H^1(\mathbb{Q}, E) \to \prod_{p \le \infty} H^1(\mathbb{Q}_p, E)\}$$

  It is expected to be finite (contained in refined BSD). If finite, $Ш(E) \cong N \oplus N$ (by Cassels–Tate pairing).

Closely related are Selmer groups. For $P \in E(\mathbb{Q})$ and $Q \in E(\overline{\mathbb{Q}})$ such that $mQ = P$, for $\sigma \in G_{\mathbb{Q}}$, $\sigma Q - Q \in E[m]$. Thus this gives a 1-cocycle in Galois cohomology:

$$E(\mathbb{Q}) \to H^1(\mathbb{Q}, E[m])$$

This gives an injection $\frac{E(\mathbb{Q})}{mE(\mathbb{Q})} \hookrightarrow H^1(\mathbb{Q}, E[m])$. To prove weak Mordell–Weil, we use the commutative diagram

$$\begin{tikzcd}
\frac{E(\mathbb{Q})}{mE(\mathbb{Q})} \arrow[r, hook] \arrow[d] & H^1(\mathbb{Q}, E[m]) \arrow[d] \\
\frac{E(\mathbb{Q}_p)}{mE(\mathbb{Q}_p)} \arrow[r, hook] & H^1(\mathbb{Q}_p, E[m])
\end{tikzcd}$$

This is helpful because if $p \nmid m$ and E has good reduction at p, then the local Galois cohomology classes land in **unramified classes** $H^1_{\mathrm{nr}}(\mathbb{Q}_p, E[m])$. So, the image of $E(\mathbb{Q})/mE(\mathbb{Q})$ lies in $\mathrm{Sel}_m(E) \subset H^1(\mathbb{Q}_S/\mathbb{Q}, E[m])$ (S is the set of bad primes), where the Selmer group is the best you could get by intersecting all preimages of images of $\frac{E(\mathbb{Q}_p)}{mE(\mathbb{Q}_p)} \to H^1(\mathbb{Q}_p, E[m])$, and $H^1(\mathbb{Q}_S/\mathbb{Q}, E[m])$ is finite by trivial reasons (finitely many number fields of bounded ramification). This gives rise to fundamental exact sequence

$$0 \to E(\mathbb{Q})/mE(\mathbb{Q}) \to \mathrm{Sel}_m(E) \to Ш(E)[m] \to 0$$

Aside: if $m = p$, we expect

$$\#Ш(E)[p] = p^{2k}$$

We know $E(\mathbb{Q})/pE(\mathbb{Q}) = (\mathbb{Z}/p\mathbb{Z})^{r_E + \dim_{\mathbb{F}_p} E[p](\mathbb{Q})}$. In particular, if $E[p](\mathbb{Q}) = 0$ and $\mathrm{Sel}_p(E) = \mathbb{Z}/p\mathbb{Z}$, then the expectation implies $r_E = 1$.

If we take $\varinjlim_n$ on the exact sequence with $m = p^n$, then

$$0 \to E(\mathbb{Q}) \otimes \mathbb{Q}_p/\mathbb{Z}_p (\cong (\mathbb{Q}_p/\mathbb{Z}_p)^{r_E}) \to \mathrm{Sel}_{p^\infty}(E) \to Ш(E)[p^\infty] \to 0.$$

So finiteness of Sha implies

$$\mathrm{cork}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(E) = r_E.$$

We have a considerable progress on unconditional

$$\mathrm{cork}_{\mathbb{Z}_p}(E) = 1 \Rightarrow r_E = 1.$$

Now we talk about results.

Theorem 1.1 (Coates–Wiles, Rubin)**.** *If $E/\mathbb{Q}$ has CM, then*

$$L(1, E) \neq 0 \Rightarrow r_E = 0 \textit{ (Coates–Wiles)}, \#Ш(E) < \infty \textit{ (Rubin)}$$

The proofs exploit the arithmetic of CM fields, especially **elliptic units**, explicitly defined units in ring class fields of CM fields. This is the first use of Iwasawa theory to prove something nontrivial about elliptic curves.

This is a more well-advertised direction of results on BSD; the converse direction is usually referred as **Converse Theorem**.

Theorem 1.2 (Rubin, "Converse to Coates–Wiles")**.** *If $E/\mathbb{Q}$ has CM, then*

$$\mathrm{cork}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(E) = 0 \Rightarrow L(1, E) \neq 0$$

Note $r_E = 0, \#Ш(E) < \infty$ implies the condition. Combined with the converse direction, knowing p^∞-Selmer group gives the result about whole Sha.

Next up: Gross–Zagier, "converse to Gross–Zagier", and very recent theorems (in the course of being written down) and their proofs.

Thursday number theory seminar: 4:30 Thursday, hopefully.

Working seminar: thesis works...

2. Lecture 02, 9/2

We will talk about Gross–Zagier. Let $E/\mathbb{Q}$ with a conductor N. Then there is a nonconstant map $\phi : X_0(N) \to E$ such that $\phi(\infty) = 0$. Such a map is not unique as one can compose this with an isogeny of E, but other than that it's mostly unique.

Choose an imaginary quadratic field such that all $\ell \mid N$ split in K. We can then choose an ideal $\mathcal{N} \subset \mathcal{O}_K$ such that $\mathcal{O}_K/\mathcal{N} \cong \mathbb{Z}/N\mathbb{Z}$. Consider a cyclic isogeny

$$y = [\mathbb{C}/\mathcal{O}_K \twoheadrightarrow \mathbb{C}/\mathcal{N}^{-1}] \in X_0(N)(\mathbb{C})$$

This point is actually defined over H, the Hilbert class field of K. Thus, $\phi(y) \in E(H)$. A **Heegner point** is

$$y_K = \sum_{\sigma \in \mathrm{Gal}(H/K)} \sigma \cdot \phi(y) \in E(K).$$

Theorem 2.1 (Gross–Zagier)**.**

$$L'(1, E/K) = c\langle y_K, y_K \rangle_{\mathrm{NT}}$$

where c is an explicit constant.

Remark 2.1. Note that $L(s, E/K) = L(s, E)L(s, E^K) = L(s, E)L(s, E \otimes \chi_K)$. Also that $L(s, E)$ around $s = 1$ makes sense uses modularity.

From the Heegner hypothesis, the root number $w(E/K)$ is -1 (from nonarchimedean factors +1, from the archimedean factor -1). So $L(1, E/K) = 0$. From Gross–Zagier, $L'(1, E/K) \neq 0$ iff y_K is of infinite order.

If y_K is torsion, then $L(s, E/K)$ vanishes up to order at least **three** at $s = 1$. Goldfeld used this to give a class number formula for imaginary quadratic field.

Theorem 2.2 (Kolyvagin)**.** *Assumptions as above. If y_K is not torsion, then $\mathrm{rk}\, E(K) = 1$ and $\#Ш(E/K) < \infty$. From this,*

$$\mathrm{rk}\, E(\mathbb{Q}) = \frac{1 - w_{E/\mathbb{Q}}}{2}, \quad \mathrm{rk}\, E^K(\mathbb{Q}) = \frac{1 + w_{E/\mathbb{Q}}}{2}$$

Also in fact $Ш(E/\mathbb{Q})$ and $Ш(E^K/\mathbb{Q})$ are both finite.

This used the method of Euler systems. Combining this with Gross–Zagier and some analytic number theory input, we get

Theorem 2.3 (Gross–Zagier, Kolyvagin, Iwaniec, Bump–Friedberg–Hoffstein)**.** *For an elliptic curve $E/\mathbb{Q}$,*

$$\mathrm{ord}_{s=1} L(s, E) = 1 \Rightarrow \mathrm{rk}\, E(\mathbb{Q}) = 1, \#Ш(E/\mathbb{Q}) < \infty$$

$$\mathrm{ord}_{s=1} L(s, E) = 0 \Rightarrow \mathrm{rk}\, E(\mathbb{Q}) = 0, \#Ш(E/\mathbb{Q}) < \infty$$

Remark 2.2. Actually one gets

$$\#Ш(E/K) \leq (*)[E(K) : \mathbb{Z}y_K]^2$$

where $(*)$ is an explicit constant. This is expected from refined BSD because $[E(K) : \mathbb{Z}y_K]^2 = \langle y_K, y_K \rangle_{\mathrm{NT}} / \operatorname{reg} E_K$.

Remark 2.3. Here, analytic number theory input is about finding a good imaginary quadratic field such that $\operatorname{ord}_{s=1} L(s, E) = 1$ and also $\operatorname{ord}_{s=1} L(s, E/K) = 1$, which means $\operatorname{ord}_{s=1} L(s, E \otimes \chi_K) = 0$. So one wants to find K that satisfies Heegner hypothesis and also that $L(1, E \otimes \chi_K) \neq 0$.

Perrin-Riou proved a p-adic version of Gross–Zagier and Kolyvagin. Kato then gave an alternate proof of rank 0 part of the above theorem without using Heegner points but using Iwasawa theory. More precisely, Kato constructed motivic classes in

$$z_{Kato} \in H^1(\mathbb{Q}, T_p E)$$

which is unramified away from p.

Remark 2.4. This is a geometric realization of Rankin–Selberg convolution against Eisenstein series,

$$L(s, f \times E_1(\chi, \psi)) = L(s, f, \chi) L(s, f, \psi)$$

To sketch how they're actually constructed, consider the Kummer map

$$\mathcal{O}(Y_1(N))^\times \to H^1_{\text{ét}}(Y, \mathbb{Z}_p(1))$$

Then modular units are sent to classes whose Hecke eigensystems look like weight 2 Eisenstein series. Then consider taking a cup product of two such classes, landing in $H^2_{\text{ét}}(Y, \mathbb{Z}_p(2))$. It goes to 0 if you send to $H^2_{\text{ét}}(Y_{\overline{\mathbb{Q}}}, \mathbb{Z}_p(2))$ (we know what Hecke eigensystem could appear there), so it should come from a class in $H^1(\mathbb{Q}, H^1_{\text{ét}}(Y_{\overline{\mathbb{Q}}}, \mathbb{Z}_p(2)))$. From this one gets a de Rham class and geometric realization of Rankin–Selberg.

At p, consider the image of z_{Kato} through

$$H^1(\mathbb{Q}, T_p E) \xrightarrow{\operatorname{loc}_p} \frac{H^1(\mathbb{Q}_p, T_p E)}{H^1_f(\mathbb{Q}_p, T_p E)}.$$

Here in this case $H^1_f(\mathbb{Q}_p, T_p E)$ is the image of local points $E(\mathbb{Q}_p) \otimes \mathbb{Z}_p$ via Kummer map. If z_{Kato} came from a global point, then $\operatorname{loc}_p(z_{Kato})$ would be in H^1_f. Composing with the Bloch–Kato dual exponential map

$$\exp^* : \frac{H^1(\mathbb{Q}_p, T_p E)}{H^1_f(\mathbb{Q}_p, T_p E)} \hookrightarrow \mathbb{Q}_p$$

one gets a formula, **the explicit reciprocity law**,

$$\exp^*(\operatorname{loc}_p(z_{Kato})) = \frac{L(1, E)}{\Omega_E} \in \mathbb{Q} \subset \mathbb{Q}_p$$

This is consistent with our expectation because if $L(1, E) \neq 0$ then we don't expect many global points and in particular it would be weird if we have a global class that seems like it comes from a global point. Using Euler systems of such special classes, Kato proved the rank 0 case of the above theorem.

If $L(1,E)=0$, then $\mathrm{loc}_p(z_{Kato}) \in H^1_f$. Is it related to the Kummer image of a Heegner point? If $\mathrm{loc}_p(z_{Kato}) \neq 0$, what does that mean? We will see that if so, $L(1,E)$ vanishes up to order exactly one, and that Heegner point is not torsion, etc. This is a general theme of **Converse theorems**.

Theorem 2.4 (Skinner–Urban, Kato)**.** *Assume*

- *$E[p]$ is irreducible as a $G_{\mathbb{Q}}$-module,*
- *E has good ordinary or multiplicative reduction at p,*
- *and there is $\ell \| N$, $\ell \neq p$, such that $E[p]$ is ramified at ℓ.*

Then

$$\mathrm{cork}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(\mathbb{E}/\mathbb{Q}) = 0 \Rightarrow L(1,E) \neq 0.$$

Many of these hypotheses can be weakened.

Remark 2.5. The third hypothesis rules out elliptic curves with complex multiplication. This is not a problem because complex multiplication case is known by Rubin's theorem.

Using BDP, Skinner proved

$$\mathrm{cork}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(E/\mathbb{Q}) = 1 \Rightarrow \mathrm{ord}_{s=1} L(s,E) = 1$$

with additional hypothesis that a global class in Selmer group has nonzero image via $\mathrm{loc}_p : \mathrm{Sel}_{p^\infty}(E/\mathbb{Q}) \twoheadrightarrow E(\mathbb{Q}_p) \otimes \mathbb{Q}_p/\mathbb{Z}_p$. The hypothesis could be now removed by subsequent development and replaced by hypotheses that look like those of the rank 0 Converse theorem. As mentioned above, such hypotheses exclude CM elliptic curves, and the results of Burungale–Tian, Burungale–Tian–Skinner cover similar results for CM ellipitic curves (that use Iwasawa theory of Heegner points as well as that of CM fields).

Going forward, we will talk about:

- a rank 1 converse theorem for CM curves,
- a proof of "Perrin–Riou's conjecture" and its consequences, namely

$$0 \neq \mathrm{loc}_p z_{Kato} \in H^1_f(\mathbb{Q}_p, E) \Rightarrow \mathrm{ord}_{s=1} L(s,E) = 1$$

  and consequences like giving a bound for $\#Ш$.

3. Lecture 03, 9/7

To prove the rank one converse theorem, one might try to prove that Heegner points are nontrivial, the so-called **Heegner point main conjecture**. By Iwasawa theory, the Heegner point main conjecture is equivalent to another main conjecture, the **BDP main conjecture**. The BDP main conjecture is more amenable to approach using either arithmetic of CM fields or automorphic approach.

Recall the setting of Heegner points. Consider an elliptic curve $E/\mathbb{Q}$ of conductor N, and a modular parametrization $X_0(N) \xrightarrow{\phi} E$ where $\infty \mapsto 0$. Let K be a imaginary quadratic field satisfying the Heegner hypothesis. For $(m, ND_K) = 1$, let $\mathcal{O}_m = \mathbb{Z} + m\mathcal{O}_K$ be an order of K. There is an ideal $\mathcal{N}_m \subset \mathcal{O}_m$ such that $\mathcal{O}_m/\mathcal{N}_m \cong \mathbb{Z}/N\mathbb{Z}$. The cyclic isogeny $[\mathbb{C}/\mathcal{O}_m \to \mathcal{O}/\mathcal{N}_m^{-1}]$ gives rise to a $K[m]$-point $P[m]$ of $X_0(N)$, where $K[m]$ is the ring class field characterized by

$$\mathrm{Gal}(K[m]/K) \cong K^\times \backslash \mathbb{A}^\times_{K,f} / \widehat{\mathcal{O}}^\times_m$$

Let $y[m] \in E(K[m])$ be $\phi(P[m])$. We get a compatible family of rational points, something like

$$N_{K[ml]/K[m]}P[ml] = \text{expressed in terms of } P[m] \text{ and Hecke operators}$$

Remark 3.1. $K[m]/K$ is **anticyclotomic**. Namely, the complex conjugation τ acts on $\sigma \mapsto \sigma^{-1}$ on $\mathrm{Gal}(K[m]/K)$. The maximal $\mathbb{Z}_p^d$-extension $\mathcal{K}_\infty/K$ is $d = 2$ which is unramified away from p, and $\mathcal{K}_\infty = K_\infty K_\infty^{\mathrm{cyc}}$ where K_∞ is the **anticyclotomic $\mathbb{Z}_p$-extension** and K_∞^{cyc} is the cyclotomic extension, and K_∞ is of finite index inside $\cup_m K[p^m]$.

Let $K_n \subset K_\infty$ be such that $\mathrm{Gal}(K_n/K) \cong \mathbb{Z}/p^n\mathbb{Z} := \Gamma_n$. Then $K_n \subset K[p^{d(n)}]$. Let $y_n = N_{K[p^{d(n)}]/K_n}(y[p^{d(n)}])$ and $\mathcal{H}_n = \langle y_m \mid m \leq n\rangle \subset E(K_n) \otimes \mathbb{Z}_p$, which has $\mathbb{Z}_p[\Gamma_n]$-action. If $n \gg 0$, then it's very big (Cornut, Vatsal, using ergodic theory).

We have a Kummer map $E(K_n) \otimes \mathbb{Z}_p \hookrightarrow H^1_f(K_n, T_pE) = \varprojlim_r \mathrm{Sel}_{p^r}(E/K_n)$. Inside it we have $\mathcal{H}_n$. Taking inverse limit with respect to n, we have

$$\mathcal{H}_\infty = \varprojlim_n \mathcal{H}_n \subset \varprojlim_n H^1_f(K_n, T_pE) = H^1_f(K_\infty, T_pE).$$

Here inverse limit is really taken using corestriction(=norm) maps. On these, we have an action of $\varprojlim_n \mathbb{Z}_p[\Gamma_n] = \mathbb{Z}_p[[\Gamma]]$.

Assume G_K acts irreducibly on $E[p]$ and $p \nmid N$ is a good ordinary prime. Then $H^1_f(K_\infty, T_pE)$ is a free Λ-module of rank 1. Also $\mathcal{H}_\infty$ is a free Λ-module of rank 1. The argument goes:

- By Cornut–Vatsal, $\mathcal{H}_\infty$ is a free Λ-module of rank 1 ("there is a nontrivial Heegner point").
- Using the above bullet point, we bound the rank of Selmer group (Howard, using Kolyvagin).

Let $\mathrm{Sel}_{p^\infty}(E/K_\infty) = \varinjlim_n \mathrm{Sel}_{p^n}(E/Kn)$. Then its Pontryagin dual X is a finitely generated Λ-module. Recall that there is a structure theorem for finitely generated Λ-modules, namely any such module is pseudo-isomorphic to finite sum of cyclic Λ-modules (i.e. those of form $\Lambda/(f)$). This means that there is a map between them with pseudo-null (=localization at height 1 primes is zero=finite order) kernel and cokernel. Using this we define the characteristic ideal to be the product of f's appearing in the structure theory. In our context, we have

Theorem 3.1 (Howard).
- *$X \sim \Lambda \oplus M \oplus M$ for a finitely generated torsion Λ-module M,*
- $\mathrm{ch}_\Lambda(M) \mid \mathrm{ch}_\Lambda\left(\frac{H^1_f(K_\infty, T_pE)}{\mathcal{H}_\infty}\right)$

Conjecture 3.1 (Heegner point main conjecture). $\mathrm{ch}_\Lambda(M) = \mathrm{ch}_\Lambda\left(\frac{H^1_f(K_\infty, T_pE)}{\mathcal{H}_\infty}\right)$.

What about $\mathrm{Sel}_{p^\infty}(E/K)$? It's $\mathrm{Sel}_{p^\infty}(E/K_\infty)[\gamma - 1]$ for $\gamma \in \Gamma$ a topological generator. So, if we assume that the corank of it is 1, then from above we get that $M/(\gamma - 1)M$ is finite. This equivalent to that $(\gamma - 1)$ does not divide $\mathrm{ch}_\Lambda(M)$. Heegner point main conjecture says that $\gamma - 1$ should then not divide $\mathrm{ch}_\Lambda\left(\frac{H^1_f(K_\infty, T_pE)}{\mathcal{H}_\infty}\right)$ which is equivalent to that the image of $\mathcal{H}_\infty$ modulo $\gamma - 1$ in $H^1_f(K_\infty, T_pE)$ modulo $\gamma - 1$ has finite index. This implies that y_K is of infinite order; this is because $\mathcal{H}_\infty = \Lambda(\varprojlim y_m)$. We have thus shown that

$$\text{Heegner point main conjecture} \quad \Rightarrow \quad \text{Converse to Gross–Zagier, Kolyvagin}$$

We are lying a little bit, because the Converse theorem assumes that the corank of Selmer group over $\mathbb{Q}$ is 1, not over K. To go to $\mathrm{Sel}_{p^\infty}(E/\mathbb{Q})$, we want K so that $\mathrm{Sel}_{p^\infty}(E^K/\mathbb{Q})$ is finite. This with Heegner hypothesis forces $w(E/\mathbb{Q}) = -1$, but this can be disregarded by the proof of

Parity conjecture in this case (Dokchitzer, Nekovar), that $w(E/\mathbb{Q})$ and corank of Selmer group $E/\mathbb{Q}$ have the same parity.

We now talk about Selmer groups. We have

$$H^1(K_\infty, T_pE) = \varprojlim_n H^1(K_n, T_pE) = \varprojlim_n H^1(K, T_pE \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[\Gamma_n]) = H^1(K, T_pE \otimes \Lambda)$$

$$\mathrm{Sel}_{p^\infty}(E/K_\infty) = \varinjlim_r \mathrm{Sel}_{p^r}(E/K_\infty) = \varinjlim_r \varinjlim_n H^1(K_n, E[p^r]) = \varinjlim_r \varinjlim_n H^1(K, E[p^r] \otimes \mathrm{Hom}(\mathbb{Z}_p[\Gamma_n], \mathbb{Z}/p^r))$$

$$= \varinjlim_r H^1(K, E[p^r] \otimes \mathrm{Hom}(\Lambda, \mathbb{Z}/p^r)) = H^1(K, T_pE \otimes \mathrm{Hom}(\Lambda, \mathbb{Q}_p/\mathbb{Z}_p))$$

4. Lecture 04, 9/9

We will talk about Selmer groups. Let M be a module with a topological action of G_F for a number field F, which is unramified away from finitely many primes of F. Then given a **Selmer structure** $\mathcal{F} = (\mathcal{F}_v)_v$, which is just $\mathcal{F}_v \subset H^1(K_v, M)$, with the condition that $\mathcal{F}_v = H^1_{\mathrm{nr}}(K_v, M)$, we define the Selmer group $H^1_{\mathcal{F}}(F, M)$ to be

$$H^1_{\mathcal{F}}(F, M) = \{c \in H^1(F, M) \mid \mathrm{loc}_v(c) \in \mathcal{F}_v\}$$

For example, for $M = E[p^n]$ with $\mathcal{F}_v$ being the image of $E(F_v)/p^nE(F_v)$, then this Selmer group $H^1_{\mathcal{F}}(F, E[p^n])$ is exactly the $\mathrm{Sel}_{p^n}(E/F)$ we defined before.

Another example is the so-called **Bloch–Kato Selmer group**. Let V be a p-adic Galois representation, over $L/\mathbb{Q}_p$. Then we define

$$\mathcal{F}_v = \begin{cases} H^1_{\mathrm{nr}}(F_v, V) & v \nmid p \\ H^1_f(F_v, V) & v | p \end{cases}$$

where $H^1_f(F_v, V) = \ker(H^1(F_v, V) \to H^1(F_v, V \otimes_{\mathbb{Q}_p} B_{\mathrm{cris}}))$. Considering a global H^1 class as an extension of $L[G_F]$-modules $0 \to V \to X \to L \to 0$, the condition of being in the Bloch–Kato Selmer group means that the extension splits over I_v for all $v \nmid p$, and that the extension splits over F_v after tensoring with B_{cris} for all $v|p$. This menas that the extension is "as crystalline as possible." This means just that

$$0 \to D_{\mathrm{cris}}(V) \to D_{\mathrm{cris}}(X) \to D_{\mathrm{cris}}(L) \to 0$$

is exact.

Now suppose you have an integral Galois representation $T \subset V$, a G_F-stable $\mathcal{O}_L$-lattice. Then we define $H^1_f(F_v, T)$ to be the preimage of $H^1_f(F_v, V)$ via $H^1(F_v, T) \to H^1(F_v, V)$. This is a little weird because by definition $H^1_f(F_v, T)$ contains all of $H^1(F_v, T)_{\mathrm{tors}}$. For example, $H^1_f(F_v, T_pE)$ is the image of $E(F_v)\widehat{\otimes}\mathbb{Z}_p$, then if v does not divide p, then this kills all but p-torsion part of $E(F_v)$.

On the other hand, we can consider $A = V/T$ and its Galois cohomology. We define $H^1_f(F_v, A)$ to be the image of $H^1_f(F_v, V)$ in $H^1(F_v, A)$. This has an opposite behavior that it might be smaller than one might want. For example, $H^1_f(F, E[p^\infty])$, where $E[p^\infty] = V_pE/T_pE$, is equal to $E(F_v) \otimes \mathbb{Q}_p/\mathbb{Z}_p$.

The Bloch–Kato Selmer group is good because you only need Galois representation and not geometry.

Example 4.1. Suppoes $F = \mathbb{Q}$ and $V = \mathbb{Q}_p(m) \supset T = \mathbb{Z}_p(m)$. Then

$$H^1_f(\mathbb{Q}_\ell, V) = H^1_{\mathrm{nr}}(\mathbb{Q}_\ell, V) = V/(\mathrm{Frob}_\ell - 1)V = V/(\ell^m - 1)V,$$

which is zero if $m \neq 0$. On the other hand,

$$H^1_f(\mathbb{Q}_p, V) = \begin{cases} H^1(\mathbb{Q}_p, V) & m > 1 \\ 0 & m < 1 \\ \mathbb{Q}_p \times (\mathbb{Z}_p^\times \widehat{\otimes} \mathbb{Q}_p) \subset \varprojlim_m (\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^{p^m}) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p = H^1(\mathbb{Q}_p, \mathbb{Q}_p(1)) & m = 1 \end{cases}$$

Conjecturally, given a geometric pure V, there is $L(s, V) = \prod_v L_v(q_v^{-s})$ where $L_v(X) = \det(1 - X \operatorname{Frob}_v^{-1}|_{V_{I_v}})$, which is convergent for $\mathrm{Re}(s) \gg 0$, has analytic continuation, Weil numbers, blah blah... **Note that we are using geometric Frobenius here**.

The rank part of BSD in this case would be stated as

$$\dim_L H^1_f(F, V) - \dim_L H^0_f(F, V) = \mathrm{ord}_{s=0} L(s, V^\vee(1))$$

where $V^\vee(1) = \mathrm{Hom}_{\mathrm{cts}}(V, \mathbb{Q}_p/\mathbb{Z}_p(1))$. This translates into our usual BSD for an elliptic curve E by taking $V = V_p(E)$, because

$$L(s, V^\vee(1)) = L(s+1, V^\vee) = L_{arith}(s+1, V) = L(s+1, E)$$

so it translates into $\mathrm{ord}_{s=1} L(s, E)$.

Example 4.2. Given an algebraic Hecke character $\psi : F^\times \backslash \mathbb{A}_F^\times | rar \mathbb{C}^\times$, which means that $\psi_\infty|_{(F\otimes\mathbb{R})_0^\times}$ is an algebraic character of $(F \otimes \mathbb{C})^\times$ restricted to $(F \otimes \mathbb{R})_0^\times$. An algebraic character of $(F \otimes \mathbb{C})^\times$ means it is of form $\prod_{\tau : F \hookrightarrow \mathbb{C}} z_\tau^{n_\tau}$ for $n_\tau \in \mathbb{Z}$. Given such character, we can give a p-adic character of G_F using class field theory. Recall that we have Artin reciprocity

$$\mathrm{rec}_F : F^\times \backslash \mathbb{A}_F^\times / (F \otimes \mathbb{R})_0^\times \to G_F^{\mathrm{ab}}$$

So giving a character $\sigma_\psi : G_F \to G_F^{\mathrm{ab}} \to \overline{\mathbb{Q}}_p^\times$ is the same as giving a character $\mathbb{A}_F^\times \to \overline{\mathbb{Q}}_p^\times$ which is trivial over $F^\times$ and $(F \otimes \mathbb{R})_0^\times$. Because ψ is algebraic, there is ρ an algebraic character $(x_\tau) \mapsto \prod x_\tau^{n_\tau}$. Then $\psi(x)\rho^{-1}(x_\infty)$ is trivial over archimedean part, but it is not trivial over $F^\times$. So ultimately we take

$$\sigma_\psi = \psi(x)\rho^{-1}(x_\infty)\rho(x_p)$$

where we fix an identification $\overline{\mathbb{Q}}_p \cong \mathbb{C}$ so thus $(F \otimes \overline{\mathbb{Q}}_p)^\times \cong (F \otimes \mathbb{C})^\times$. One can show that this is even a continuous character.

In general what we mean by Iwasawa theory is that V is varied in a p-adic family. Algebraic part is about varying Selmer groups and analytic part is about varying L-values.

Example 4.3. Consider the cyclotomic $\mathbb{Z}_p$-extension $\mathbb{Q}_\infty/\mathbb{Q}$. Let $1+p \in \Gamma$ be denoted as $\gamma \in \Lambda = \mathbb{Z}_p[[\Gamma]]$ (the group element, not the coefficient). Consider the canonical character $\Psi : G_\mathbb{Q} \twoheadrightarrow \Gamma \hookrightarrow \Lambda^\times$. This can be thought as a family of $\mathbb{Z}_p(\omega^{-m}\epsilon^m)$, in that Ψ-action on $\Lambda/((1+T) - \epsilon(\gamma)^m)\Lambda$ is as $G_\mathbb{Q}$-representation $\mathbb{Z}_p(\omega^{-m}\epsilon^m)$. Given any continuous character $\Gamma \to \mathbb{Q}_p^\times$, one obtains a continuous homomorphism $\Lambda \to \mathbb{Q}_p$ sending $T \mapsto$ (image of γ)-1.

Question: is

$$H^1_{\mathcal{F}}(F, T_pE \otimes \Lambda)/(1 + T - \epsilon^m(\gamma)) = H^1_f(F, T_pE(\omega^{-m}\epsilon^m))?$$

Answer is not simple, depending on Hodge–Tate weights. This is a work of Greenberg.

5. Lecture 05, 9/14

Let K be an imaginary quadratic field. Suppose $p = v\overline{V}$. Choose embedddings $K \hookrightarrow \overline{\mathbb{Q}}$, $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}, \overline{\mathbb{Q}}_p$ so that the p-adic embedding induces v. Consider a Hecke character $\psi : \mathbb{A}_K^\times \to \mathbb{C}^\times$. Suppose it is algebraic, in a way that $\psi_\infty(x_\infty) = x_\infty^a \overline{x}_\infty^b$ (here $x_\infty \in \mathbb{C}$ from $x \in \mathbb{A}_K^\times$ is determined by complex embedding.

Then, $\psi(x) x_\infty^{-a} \overline{x}_\infty^{-b} \in \overline{\mathbb{Q}}$, and is actually in some finite extension $\overline{\mathbb{Q}} \supset F/K$. The p-adic embedding of $\mathbb{Q}$ induces $\lambda \mid p$. Then we define the character of G_K, σ_ψ, as

$$\sigma_\psi(g) = \psi(x) x_\infty^{-a} \overline{x}_\infty^{-b} x_v^a x_{\overline{v}}^b$$

where $g \in G_K^{\mathrm{ab}}$ corresponds to an idele x via class field theory.

In this normalization, the Hodge–Tate weight at v is a and the Hodge–Tate weight at $\overline{v}$ is b.

Example 5.1. If $E/\mathbb{Q}$ has a CM by K, then V_p has an action of $(K \otimes_{\mathbb{Z}} \mathbb{Q}_p)^\times = K_v^\times \times K_{\overline{v}}^\times$. Thus, there is a decomposition $V_p E = V_v E \oplus V_{\overline{v}} E$ where each factor is one-dimensional over $\mathbb{Q}_p$ and has an action of G_K (note that CM is defined over K, even though elliptic curve is defined over $\mathbb{Q}$). The Galois representation appearing at $V_v E$ is σ_ψ where $\psi_\infty(z) = z^{-1}$.

Given a Hecke character of infinity type (a, b), namely $\psi_\infty(z) = z^a \overline{z}^b$, if $a < 0 \leq b$, it is **critical**, and $n \in \mathbb{Z}$ is a **critical value** for $L(S, \psi)$ if $a + n \leq 0 < b + n$ or $a + n > 0 \geq b + n$.

In this case, what is the Bloch–Kato Selmer group $H_f^1(K_w, \sigma_\psi)$ where σ_ψ is $F_\lambda(\sigma_\psi)$?

- If $w \nmid p$, it's 0.
- If $w = v$,
  - if $a < 0$, $H^1(K_v, \sigma_\psi)$,
  - if $a \geq 0$, it's 0.
- If $w = \overline{v}$, it's similar.

What happens when we vary ψ p-adically? Let K_∞ be the $\mathbb{Z}_p^2$-extension of K. Let $\Gamma_K = \mathrm{Gal}(K_\infty/K)$. Let $\Lambda = \mathbb{Z}_p[[\Gamma_K]]$. Let $\Psi : G_K \twoheadrightarrow \Gamma_K \subset \Lambda^\times$. Then we can extend this to $G_K \to (\mathcal{O}_{F_\lambda} \otimes_{\mathbb{Z}_p} \Lambda)^\times = \Lambda_{\mathcal{O}}^\times$, defined as $\sigma_\psi \otimes \Psi$.

Let $\chi : \Gamma_K \to \overline{\mathbb{Q}}_p$ be a character. Note rec_K sends $(\mathcal{O}_K \otimes \mathbb{Z}_p)^\times = \mathcal{O}_v^\times \oplus \mathcal{O}_{\overline{v}}^\times$ to Γ_K with finite index image, and say χ restricted to a finite index subgroup of $\mathcal{O}_v^\times$ is $u \mapsto u^c$ and similarly $u \mapsto u^d$ for $\mathcal{O}_{\overline{v}}^\times$. Extend this linearly, and consider

$$G_K \xrightarrow{\sigma_\psi \otimes \Psi} \Lambda_{\mathcal{O}}^\times \xrightarrow{\chi} \overline{\mathbb{Q}}_p^\times$$

which can be called as $\sigma_\psi \chi$. This Galois representation is the Galois representation corresponding to the Hecke character ψ_χ which is of infinity type $(a+c, b+d)$. Thus, based on (c, d) the infinity type can change as much as we want, and in particular we cannot impose the Bloch–Kato Selmer group condition consistent over the whole range of interpolation; it is bound to work for only a certain range. This is actually a universal phenomenon and something that could be exploited.

Consider a class $c \in H^1(K, \mathcal{O}(\sigma_\psi) \otimes \Lambda^\times)$ so that $\mathrm{loc}_w(c)$ is unramified when w does not divide p, $\mathrm{loc}_v(c)$ to be anything, and $\mathrm{loc}_{\overline{v}}(c)$ be unramified. This defines a Selmer group $\mathrm{Sel}(\psi/K_\infty)$. Let $X(\psi/K_\infty)$ be the Pontryagin dual of $\mathrm{Sel}(\psi/K_\infty)$. We have

$$\mathrm{Sel}(\psi/K_\infty)[\chi] \subset H^1(K, \mathcal{O}(\sigma_\psi) \otimes \Lambda^*)[\chi] = H^1(K, \mathcal{O}(\sigma_\psi \chi^{-1}) \otimes \mathbb{Q}_p/\mathbb{Z}_p)$$

More generally we can take a look at $(\mathrm{Sel}(\psi/K_\infty) \otimes_{\mathcal{O}} \mathcal{O}_\psi)[\chi]$.

Then, we have something that can be called a *control theorem*, that for most χ (namely $a+c < 0 \leq b+d$),

$$\mathrm{Sel}(\psi/K_\infty)[\chi] = \text{“}\mathrm{Sel}(\psi_{\chi^{-1}}/K)\text{”}$$

Here the Selmer group at the right hand side contains the Bloch–Kato Selmer group $H^1_f(K, \sigma_{\psi_\chi} \otimes \mathbb{Q}_p/\mathbb{Z}_p)$. The difference comes from the fact that we impose classes to be 0 away from p for the Bloch–Kato Selmer group whereas Sel requires unramifiedness, and the index is in fact the product of local Tamagawa factors. This is expected because the L-value should contain the order of the Tate–Shafarevich group as well as the product of local Tamagaway factors, and we expect to show that the order of Selmer group **is** the L-value.

Because $\mathrm{Sel}(\psi/K_\infty)$ is packaging $\mathrm{Sel}(\psi_{\chi^{-1}}/K)$, and by the Bloch–Kato conjectures the order of $\mathrm{Sel}(\psi_{\chi^{-1}}/K)$ is related to $L_S(\psi\chi^{-1}, 1)$, we expect there is a p-adic packaging of L-values, which is usually called a p-**adic** L-**function**. In this case the p-adic L-function is constructe by Katz,

$$L_v^{\mathrm{Katz}}(\psi) \in \mathcal{O}^{\mathrm{nr}}[[\Gamma_K]]$$

which interpolates algebraic part of $L(\psi\chi^{-1}, 1)$. Katz constructed by constructing a p-adic interpolation of Eisenstein series and evaluating at CM points etc.

Then we can say finally what is the **Iwasawa Main Conjecture** for K is.

Conjecture 5.1 (Iwasawa Main Conjecture for K)**.**

- *$X(\psi/K_\infty)$ is a finitely generated torsion Λ-module.*
- $\mathrm{ch}_\Lambda(X(\psi/K_\infty)) = (L_v^{\mathrm{Katz}}(\psi))$.

This is a **theorem** by Rubin. This has several consequences:

- $\#X(\psi/K_\infty)_\chi = \#\,\mathrm{Sel}(\psi_{\chi^{-1}})^* = \#\,\mathrm{Sel}(\psi_{\chi^{-1}}) = \#(\Lambda_{\mathcal{O}}/\xi)_\chi$ where ξ is a generator of the characteristic ideal. This is basically $\#\mathcal{O}/L_{\mathrm{alg}}(\psi\chi^{-1}, 1)$.

Given ψ of infinty type $(-1, 0)$, we can take a CM elliptic curve E where $L(s, \psi) = L(s, E)$. That p splits means that p is ordinary for E. As $\mathrm{Sel}(\psi/K)$ contains $\mathrm{Sel}_{p^\infty}(E/\mathbb{Q})$ as a finite index subgroup, the IMC tells us that $L(1, \psi) = L(1, E)$ not vanishing is equivalent to the corank of the Selmer group being 0.

What about the rank 1 case? Conside4 ψ^c, which is of infinity type $(0, -1)$. This is not good for p-adic L-function because of the simple criticality consideration. However, if $\epsilon(\psi) = -1$ (i.e. $\epsilon(E/\mathbb{Q}) = -1$), then Rubin showed that

$$L_p^{\mathrm{Katz}}(\psi^c)(\mathbb{1}) = (*)\log_{p\text{-adic}}(\text{some Heegner point})$$

So the nontorsionness of Heegner point can be explained by nontriviality of $L_p^{\mathrm{Katz}}(\psi^c)$ at some range outside its range of interpolation. What's missing for the rank 1 BSD is about relating $\mathrm{Sel}_{p^\infty}(E/\mathbb{Q}) = \mathrm{Sel}(\psi/K_\infty)[\mathbb{1}]$ being of corank 1 and $\mathrm{Sel}(\psi^c/K_\infty)[\mathbb{1}]$ being finite.

6. Lecture 06, 9/16

**Disclaimer: I wasn't there, so the content may be more inaccurate than usual.

We compare many main conjectures. First consider the cyclotomic main conjecture; let F be a number field. Consider

$$F^\times\backslash\mathbb{A}_F^\times/\overline{F^\times_{\infty,0}\widehat{\mathcal{O}}_F^{\times(\mathfrak{p})}} = \mathrm{Cl}_F(\mathfrak{p}^\infty) \twoheadrightarrow \mathrm{Cl}_F = F^\times\backslash\mathbb{A}_F^\times/\overline{F^\infty_{\infty,0}\widehat{\mathcal{O}}_F^\times}$$

By global duality, we have an exact sequence like

$$0 \to (\text{global units}) \to (\text{local units}) \to \mathrm{Cl}_F(\mathfrak{p}^\infty) \to \mathrm{Cl}_F \to 0$$

However there are "special units" among global units. The explicit reciprocity law relates

$$\{\text{Local units}\} \,/\, \{\text{Special units}\} \xleftrightarrow{\text{explicit reciprocity law}} p\text{-adic } L\text{-function}$$

The usual main conjecture is about relating $\{\text{Local units}\} \,/\, \{\text{Special units}\}$ and $\mathrm{Cl}_F(\mathfrak{p}^\infty)$. Using the exact sequence, we can get the **Main Conjecture without L-functions**, which is about relating $\{\text{Global units}\} \,/\, \{\text{Special units}\}$ and Cl_F.

Remark 6.1. Sometimes the two end terms of the exact sequence agree for two settings. In that case it allows one to pass one Main Conjecture to the other.

In the language of Selmer groups, we have

$$\begin{tikzcd}
H^1(\mathcal{O}_F[\frac{1}{p}], \mathbb{Z}_p(1)) \arrow[r, "\mathrm{res}_v"] \arrow[dd, equal, "\text{Kummer}"'] & H^1(F_v, \mathbb{Z}_p(1)) \arrow[rr] \arrow[dd, equal] \arrow[dr, "\cong", "\text{local duality}"'] & & H^1_{\mathrm{str}^v}(\mathcal{O}_F[\frac{1}{p}], \mathbb{Q}_p/\mathbb{Z}_p)^\vee \arrow[r] & H^1_{\mathrm{str}}(\mathcal{O}_F[\frac{1}{p}], \mathbb{Q}_p/\mathbb{Z}_p)^\vee \arrow[r] & 0 \\
& & H^1(F_v, \mathbb{Q}_p/\mathbb{Z}_p)^\vee \arrow[ur, "\mathrm{res}_v^\vee"'] & & & \\
\mathcal{O}_F[\frac{1}{p}]^\times \otimes \mathbb{Z}_p & F_v^\times \otimes \mathbb{Z}_p & & & &
\end{tikzcd}$$

Here str is the Selmer condition where it is strict at all places above p, whereas str^v is the Selmer condition where it is strict at all places above p **except** v.

Another Main Conjecture: let K be an imaginary quadratic field and ψ be a Hecke character of type $(-1, 0)$. Let $\Lambda = \mathbb{Z}_p[[\Gamma_K]]$, with $p = v\overline{v}$ split. Let $\psi\psi^c = \epsilon$, the cyclotomic character. We have a similar exact sequence

$$\begin{tikzcd}
H^1(\mathcal{O}_K[\frac{1}{p}], \mathcal{O}(\psi^c)\widehat{\otimes}\Lambda) \arrow[r, "\mathrm{res}_v"] \arrow[dd, equal, "\mathscr{E}_\infty"'] & H^1(K_v, \mathcal{O}(\psi^c)\widehat{\otimes}\Lambda) \arrow[rr] \arrow[dd, equal] \arrow[dr, "\cong"] & & H^1_{\mathrm{rel,str}}(\mathcal{O}_K[\frac{1}{p}], \mathcal{O}(\psi)\widehat{\otimes}\Lambda^*)^\vee \arrow[r] \arrow[dd, equal] & H^1_{\mathrm{str,str}}(\mathcal{O}_K[\frac{1}{p}], \mathcal{O}(\psi)\widehat{\otimes}\Lambda^*)^\vee \arrow[r] & 0 \\
& & H^1(K_v, \mathcal{O}(\psi)\widehat{\otimes}\Lambda^*)^\vee \arrow[ur, "\mathrm{res}_v^\vee"'] & & & \\
\Lambda u_\psi & \Lambda & & X(\psi/K_\infty) & &
\end{tikzcd}$$

where $\mathscr{E}_\infty$ are the elliptic units. The identifications are due to Coates–Wiles, Coleman.

Theorem 6.1 (Rubin). *(1) Under the above identifiactions, $u_\psi \mapsto L_v^{\mathrm{Katz}}(\psi)$.*
(2) $\mathrm{char}(H^1(\mathcal{O}_K[\frac{1}{p}], \mathcal{O}(\psi^c)\widehat{\otimes}\Lambda)/\mathscr{E}_\infty) = \mathrm{char}(H^1_{\mathrm{str,str}}(\mathcal{O}_K[\frac{1}{p}], \mathcal{O}(\psi)\widehat{\otimes}\Lambda^*)^\vee)$.

More precisely, one divisibility is proven via Euler systems, and the equality is derived from analytic class number formula. Also, (2) is equivalent to

$$\mathrm{char}(H^1(K_v, \mathcal{O}(\psi^c)\widehat{\otimes}\Lambda)/\mathscr{E}_\infty) = \mathrm{char}(X(\psi/K_\infty))$$

which is generated by $L_v^{\mathrm{Katz}}(\psi)$ by explicit reciprocity law.

If you interchange ψ and ψ^c, you get an exact sequence

$$H^1(\mathcal{O}_K[\frac{1}{p}], \mathcal{O}(\psi^c)\widehat{\otimes}\Lambda) \xrightarrow{\mathrm{res}_{\overline{v}}} H^1(K_{\overline{v}}, \mathcal{O}(\psi^c)\widehat{\otimes}\Lambda) \to X_{\mathrm{str,rel}}(\psi/K_\infty) \to X_{\mathrm{str,str}}(\psi/K_\infty) \to 0$$

From what Rubin proved, we know that

$$\mathrm{char}(X_{\mathrm{rel,str}}(\psi^c/K_\infty) = L_p^{\mathrm{Katz}}(\psi^c)$$

Thus, if you evaluate at 1, you have

$$L_p^{\mathrm{Katz}}(\psi^c)(\mathbb{1}) \neq 0 \quad \Leftrightarrow \quad H^1_{\mathrm{rel,str}}(\mathcal{O}_K[\frac{1}{p}], \mathbb{Q}_p/\mathbb{Z}_p(\psi^c)) \text{ is finite}$$

Rubin also proved, which is later generalized by BDP, that

$$L_p^{\mathrm{Katz}}(\psi^c)(\mathbb{1}) = (*)\log_p(\text{Heegner point } P)^2$$

Thus, P is non-torsion if and only if $L'(\psi, 1) \neq 0$. In turn, $L'(\psi, 1) \neq 0$ if and only if $H^1_{\mathrm{rel,str}}(\mathcal{O}(\psi^c) \otimes \mathbb{Q}_p/\mathbb{Z}_p)$ is finite.

Finally we consider anticyclotomic Main Conjecture. Let E be any elliptic curve over $\mathbb{Q}$, K be an imaginary quadratic field satisfying Heegner hypothesis, $p \nmid N_E$, $p = v\overline{v}$ splits. Let $\Lambda^{ac} = \mathbb{Z}_p[[\Gamma_K^{ac}]]$. Then

$$H^1_{\mathrm{rel,str}}(\mathcal{O}_K[\frac{1}{p}], T_pE\widehat{\otimes}(\Lambda^{ac})^*)$$

interpolates $H^1_f(V_pE(\psi))$ for ψ with infinity type $(-n, n)$ for $n > 0$.

Remark 6.2. If ψ is of finite order, we need E to have ordinary reduction to satisfy Panchiskin condition. Thus, supersingular primes cause problems for cyclotomic twists.

Conjecture 6.1 (Main Conjecture with L-functions)**.** *(1) $X^{ac}(E/K)$ is a torsion Λ^{ac}-module.*
(2) $\mathrm{char}_{\Lambda^{ac}}(X^{ac}(E/K)) = (L_v^{ac}(E/K))$, where the p-adic L-function interpolates $L_{\mathrm{alg}}(E, \psi, 1)$.

We briefly remark about the construction of the p-adic L-function. Taking f_E as the modular form corresponding to E and θ_ψ as the theta series of ψ,

$$L(E, \psi, s) = L(f_E \times \theta_\psi, s) = L(f_E \times \theta_{\psi\|-\|^{-n}}, s + n)$$

This is better because then $\theta_{\psi\|-\|^{-n}}$ is a holomorphic modular form. Using this we see that the transcendental period is $4n$-th power of the CM period for ψ. One has two approaches, either by Hida using that it's a Rankin–Selberg L-function, or by BDP/Hsieh–Castella which generalize Katz's approach.